Amnesia by Design: On Temporal Traceability in Transformer Representations

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Abstract. Words do not preserve fixed meanings across time. `gay@1920` and `gay@1980` are different semantic objects, yet transformers treat them as the same token in the same geometric space. Masked language modeling rewards correct token prediction rather than geometric consistency across time, so temporal traceability is not directly optimized.
We introduce a controlled benchmark with known ground-truth semantic trajectories to demonstrate this limitation without corpus confounds, and formalize Timeformer, a family of token-time conditioning designs. Temporal conditioning more than doubles neighborhood drift for drifting subjects, yet no architecture matches the planted trajectories across all diagnostics. The memory-augmented extension points to a second problem: when a word develops coexisting senses, mean pooling collapses both into a centroid that represents neither.
The results matter for applications that require word meaning at a specific period, including historical semantic analysis and knowledge-intensive tasks over time-sensitive corpora. They suggest that temporal traceability requires training objectives aligned with trajectory geometry.

Keywords: Temporal semantics · Semantic change · Transformer models · Traceability · Diagnostic evaluation

1 Introduction

Words do not preserve fixed meanings across time. In 1900, *broadcast* referred to scattering seeds across a field; today it describes a social media post reaching millions of users. Whether the shift of *gay* from *cheerful* to its contemporary sense involved movement toward or away from stigmatized terms [5] is an empirical question that requires `gay@1920` and `gay@1980` as directly comparable objects in a shared geometric space.

Contextualized embeddings disambiguate senses through local sentential context, not period. A model correctly represents *virus* differently in a sentence with *malware* and *code* versus one with *infection* and *bacteria*, using the neighboring tokens as evidence. The period is irrelevant for this disambiguation. The limitation appears when one wants to query `token@time` without a supporting sentence: `gay@1920` as a directly inspectable object, not as an embedding that only separates from `gay@1980` when forced by local context. Diachronic analysis asks where a meaning moved and how close it became to other terms; it therefore needs the `token@time` object.

Masked language modeling masks a token and asks the model to reconstruct it from its neighbors. To do this well, the model learns local context; it has no direct incentive to place the same token in period-appropriate regions of the representation space. Temporal traceability means that `virus@1900` and `virus@2020` occupy different neighborhoods. It requires a training signal aligned with geometric consistency across time, which the MLM objective does not provide.

Testing this limitation requires a controlled environment where the correct answer is known. Natural-language corpora entangle semantic change with frequency, genre, and subcorpus shifts, making it hard to attribute a representational effect to architecture rather than data. The controlled benchmark we introduce isolates this effect before moving to open-domain corpora.

Against this benchmark we formalize Timeformer, a family of token-time conditioning designs. The comparison isolates the contribution of each architectural decision: whether any temporal signal helps, whether joint token-time conditioning outperforms global period injection, and whether causal prototype memory adds anything beyond conditioning. The controlled environment makes each question answerable without corpus confounders.

The paper makes four contributions: it establishes temporal traceability as a geometric property distinct from period-label recoverability; shows in a controlled controlled benchmark that standard transformer training supports this property only as a by-product; compares four token-time conditioning designs; and identifies mean-prototype aggregation as a second failure mode.

2 Related Work

Lexical semantic change detection (LSCD) is the dominant computational framing for meaning shift. Diachronic word embeddings [5] learn one distributional space per time slice, align independently trained models, and measure movement in neighborhood structure. We share the intuition that semantic change is geometric, but not the architecture: LSCD yields a scalar comparison between spaces rather than a jointly parameterized `token@time` representation. Standard benchmarks formalize this task [8, 9], while also inheriting natural-corpus confounds such as frequency and genre drift; shared tasks for Spanish show that manual annotation is required even for modern corpora [11]. Our controlled benchmark plants trajectories so that traceability can be attributed to model design rather than corpus composition.

Temporal neural language models avoid separate spaces by conditioning one model on time. Prior work uses timestamps for knowledge-intensive QA, masked language modeling, or period-stratified downstream evaluation [3, 7, 6]. These results show that temporal conditioning can improve period-appropriate prediction, but they do not test whether the same token occupies different neighborhoods at different periods in a shared space.

Linear probes provide another partial view. A period probe measures whether time is recoverable from a hidden state, not whether a token has moved to a different semantic neighborhood. We therefore report probe accuracy as auxiliary

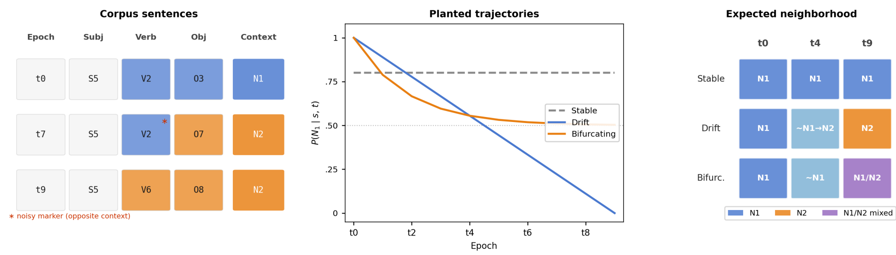

Fig. 1. Benchmark design overview. *Left*: SVO sentences for a drifting subject across three periods; an N_1 verb appearing in an N_2 sentence at $t7$ (marked $*$) illustrates the deliberate noise in local markers. *Center*: planted $P(N_1 \mid s,t)$ trajectories for Stable (flat), Drift (monotonic), and Bifurcating (levelling at 0.5) subjects across $t0$–$t9$. *Right*: expected nearest-neighbor behavior for a traceable representation.

evidence and treat neighborhood movement as the primary criterion for traceability.

Finally, semantic change may involve bifurcation rather than monotonic drift. Giulianelli et al. show that words can develop coexisting senses [4]; mean-pooled representations then collapse both senses into a centroid that represents neither. Our stable, drifting, and bifurcating trajectories test this distinction directly, and the memory extension exposes where mean-pooled prototypes fail.

3 Controlled Temporal Benchmark

The benchmark gives each subject a known temporal trajectory, making it possible to ask directly whether a model's representation of a subject at $t0$ has moved toward the expected neighborhood by $t9$. This distinguishes it from natural-language LSCD benchmarks, where ground truth is often annotated or inferred post-hoc and corpus confounds are hard to separate from genuine semantic change.

The corpus uses a subject–verb–object (SVO) grammar: every sentence consists of one subject, one verb, and one object. Two semantic neighborhoods, N_1 and N_2, partition the vocabulary of 8 verbs and 8 objects: N_1 uses $\{V1,\dots,V4\}$ and $\{O1,\dots,O4\}$; N_2 uses $\{V5,\dots,V8\}$ and $\{O5,\dots,O8\}$. A sentence generated from N_1 therefore looks like `S5 V2 O3`, while one from N_2 looks like `S5 V6 O8`; the subject token is the same in both cases, but its co-occurring words differ. Intuitively, the intended meaning of subject s at period t is how often it appears in N_1 sentences; formally, $P(N_1 \mid s,t)$ is the probability that a sentence is drawn from N_1. The vocabulary contains 30 subjects; each subject's trajectory is determined by how $P(N_1 \mid s,t)$ evolves across the 10 periods $t0$ through $t9$.

Figure 1 (center) shows the three planted trajectory classes. Stable subjects maintain an approximately constant $P(N_1 \mid s,t)$ throughout. Drift subjects follow a monotonic trajectory from near-exclusive N_1 ($P(N_1 \mid s,t) \approx 1$ at $t0$) to near-exclusive N_2 ($P(N_1 \mid s,t) \approx 0$ at $t9$): a drifting subject that appears in sentences like `S5 V2 O3` early on will appear in sentences like `S5 V6 O8` by the final period. Bifurcating subjects begin in N_1 and move toward a mixed state ($P(N_1 \mid s, t) \approx 0.5$) where both neighborhoods appear with equal frequency, approximating two coexisting senses. The benchmark contains 10 subjects per class.

Each local marker is drawn from the opposite neighborhood 25% of the time (Figure 1, left), so the model cannot recover the planted neighborhood from co-occurring words alone; it must use subject identity together with the period. The `true_context` label is withheld from training and used only for evaluation.

The synthetic benchmark isolates architectural effects that natural corpora confound with frequency, genre, and composition shifts. By planting trajectories with known $P(N_1 \mid s, t)$, we can ask directly whether a model's geometry matches the generating process. The scope of these controlled diagnostics is revisited in the Discussion.

The benchmark supports several evaluation conditions. The standard split uses the same 75% marker accuracy as training. Two harder splits force the verb, or both verb and object, to come from the opposite context. An ambiguous split sets fidelity to 50%, so local markers carry no reliable signal; only subject identity and period remain. Two additional conditions test temporal behaviour and are described in Section 5: the contrastive set presents the same sentence under two different period labels to check whether changing only the period changes the preferred context; the continuation set uses periods $t8$–$t9$, held out from MLM training (which covers $t0$–$t7$), to test generalisation.

4 Timeformer Architecture

Timeformer is designed to make a subject representation depend explicitly on the queried period. We study this through a controlled comparison: Standard has no period input, Additive injects a global period vector, Token-Time exposes token identity and period jointly before self-attention, and the Memory-Augmented extension adds causal historical prototypes before the encoder.

All models follow the same outer pipeline based on the Transformer encoder [10]. A sentence is tokenized as `[CLS] S V O [SEP]`. Each token w_i is mapped to an input embedding $z_i \in \mathbb{R}^d$. The sequence $(z_1, \dots, z_n)$ is processed by a Transformer encoder, and the final hidden state at the subject position,

$$(h_1, \dots, h_n) = \mathrm{Encoder}(z_1, \dots, z_n), \qquad h_s = h_{\mathrm{subj}},$$

is the `token@time` representation inspected by the diagnostics. All models are trained with masked language modeling (MLM, [2]): some tokens are masked, and the model reconstructs them from the remaining tokens and, when available, the period input. The variants differ only in how they build z_s before the encoder.

The temporally informed models use a shared period vector $\tau(t) \in \mathbb{R}^d$, a learned projection of sinusoidal features of t/T, where T is the number of training periods. It is distinct from the position encoding p_i, which marks subject, verb, or object position. Additive and Token-Time differ in how they combine $\tau(t)$

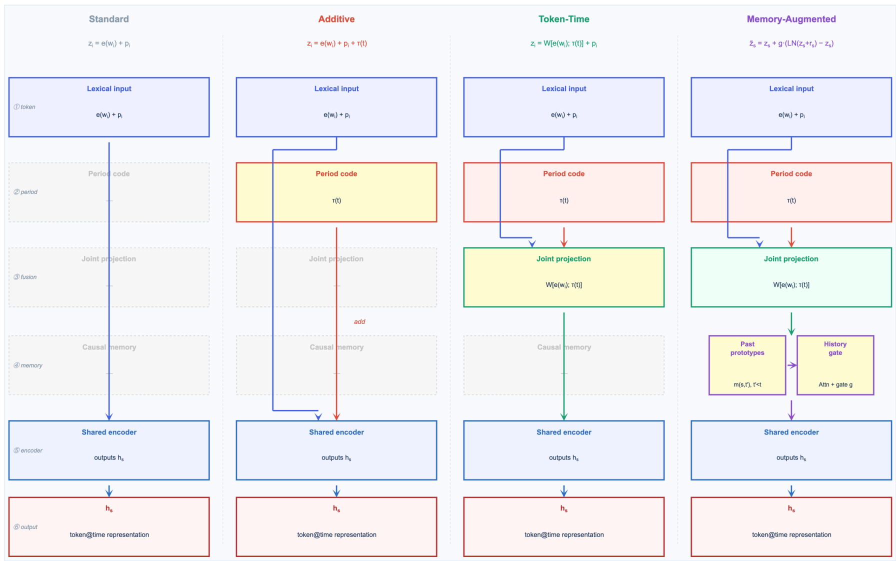

Fig. 2. Controlled design comparison. **Standard**: Transformer baseline with no temporal signal. **Additive**: temporal baseline that adds the same period vector to every token. **Token-Time**: Token-Time Transformer using joint token-time projection. **Memory-Augmented**: extension of Token-Time with Prototype Memory and Temporal Attention. New components are highlighted in yellow; dashed grey boxes mark inactive components.

with token identity; Memory-Augmented then adds causal memory. Figure 2 summarizes the comparison.

4.1 Baseline: Standard Transformer

Standard is the null hypothesis: no period signal is available, so any temporal behavior must arise from changes in the observed words alone.

$$z_i = e(w_i) + p_i.$$

4.2 Baseline: Additive Time-Conditioned Transformer

The Additive baseline asks whether knowing the current period is enough. It adds the same period vector $\tau(t)$ to every token:

$$z_i = e(w_i) + p_i + \tau(t).$$

Now `S5 V2 O3` at $t0$ and $t9$ are distinguishable: the subject receives $\tau(0)$ or $\tau(9)$. The same temporal change is also applied to the verb and object, so Additive tells the whole sentence which period it belongs to but does not explicitly bind period to each token's identity.

4.3 Token-Time Transformer

The Token-Time Transformer in Figure 2 asks whether exposing token identity and period jointly before self-attention is enough. Instead of adding the same period vector to all tokens, Token-Time concatenates each token embedding with $\tau(t)$ and projects the pair:

$$z_i = W[e(w_i);\, \tau(t)] + p_i,$$

where $e(w_i), \tau(t) \in \mathbb{R}^d$ and $W \in \mathbb{R}^{d\times 2d}$. Although W is linear, it exposes token-period inputs before self-attention and later nonlinear layers. Thus `S5` at $t0$ and `S5` at $t9$ enter the encoder as distinct period-conditioned inputs.

4.4 Memory-Augmented Extension

The Memory-Augmented extension asks whether causal history helps beyond token-time conditioning. It starts exactly like Token-Time: for a current sentence containing `S5` at $t7$, the model first builds token-time input embeddings for `[CLS] S5 V O [SEP]`. It then modifies only the subject embedding before the Transformer encoder sees the sentence.

The additional information comes from Prototype Memory. The extension stores one prototype vector per subject and past period. A prototype $m(s,t)$ is the average representation of subject s across training sentences from period t. The memory is causal: at $t7$, the model may consult $\{m(\texttt{S5},0),\ldots,m(\texttt{S5},6)\}$, but not prototypes from $t7$ or later.

Temporal attention turns this list of past prototypes into a history summary for the current subject. Let z_s be the current Token-Time subject embedding and let M_s be the matrix of available causal prototypes, one row per past period. The attention operation uses z_s as the query and M_s as keys and values:

$$r_s = \mathrm{Attn}(z_s, M_s, M_s).$$

The vector r_s summarizes the parts of the subject's history that are most relevant to the current period. A learned scalar gate g then decides how much this history summary should change the current subject embedding:

$$\tilde{z}_s = z_s + g \cdot \left(\mathrm{LN}(z_s + r_s) - z_s\right).$$

Here, LN denotes layer normalization, and g is a learned scalar gate initialized at zero so that the extension reduces to Token-Time at initialization. Finally, $\tilde{z}_s$ replaces z_s in the input sequence; verb and object embeddings are unchanged, so the subject is the only token that enters the encoder informed by causal prototype history.

5 Evaluation Protocol and Diagnostics

All models in the comparison chain are trained with masked language modeling over the synthetic SVO corpus. The planted `true_context` label is withheld from training and used exclusively for post-hoc evaluation: it tests whether the learned subject representation reflects the planted temporal trajectory. The primary evaluation target is h_s, the final hidden state at the subject position, because subjects are the lexical items whose meanings are designed to drift, remain stable, or bifurcate over time.

We use four diagnostics, each targeting a distinct aspect of temporal representation.

D1: Linear probe [1]. Asks whether h_s encodes the planted context label at all. The encoder is frozen; a linear classifier is trained on h_s representations extracted from the held-out test sentences of the standard (75%-fidelity) split and evaluated on the ambiguous and continuation splits. High accuracy indicates that period-appropriate context is linearly separable in the representation, but does not reveal whether that separation shifts systematically over time.

D2: Context drift score. Asks whether the neighborhood of a subject shifts in the expected direction across periods. For each period and subject class, we retrieve the $k = 10$ nearest neighbors of h_s by cosine similarity (excluding the query itself) and record the proportion assigned to N_1. A representation is traceable at period t if this proportion tracks $P(N_1 \mid s, t)$, the same probability that governs corpus generation at that period. For drifting subjects, the proportion should therefore decrease monotonically from $t0$ to $t9$; for stable subjects it should remain near its initial value; for bifurcating subjects it should decrease less steeply, as the representation splits between neighborhoods rather than migrating from one to the other.

D3: Contrastive diagnostic. Asks whether the period label alone drives representational change. We present the same sentence with two different period labels and compute the sign-flip rate: the proportion of pairs for which swapping the period reverses the nearest-neighbor assignment, from N_1-dominant to N_2-dominant or vice versa. A model that ignores the period input produces a sign-flip rate of zero; a model that responds to it produces a rate above zero.

D4: Continuation diagnostic. Asks whether the learned temporal mapping generalizes beyond training. We evaluate on held-out periods $t8$–$t9$, excluded from MLM training, and operationalize the diagnostic as linear-probe accuracy on representations from these periods. A model that has internalized the direction of change, rather than memorizing period-specific patterns, should preserve context recoverability in unseen periods.

The four diagnostics are interpreted through a fixed comparison chain. Additive vs. Standard isolates the effect of adding any temporal signal. Token-Time vs. Additive tests whether exposing token identity and period jointly before self-attention changes the result relative to applying the period as a uniform shift. Memory-Augmented vs. Token-Time tests the contribution of causal prototype memory, the mechanism most expected to affect D3 and D4.

For the Memory-Augmented extension, we include an *oracle-memory* diagnostic: learned prototypes are replaced by fixed vectors encoding the true historical $P(N_1 \mid s, t)$ trajectory, with the rest of the architecture unchanged. This isolates whether the failure of learned prototypes is due to the memory pathway itself or to the aggregation mechanism; improvement under oracle prototypes indicates that the pathway can use better historical signals, while learned mean prototypes remain the weak point.

6 Results

We report means and 95% confidence intervals over 31 training seeds. Temporal conditioning moves drifting subjects much more than the Standard baseline: Additive, Token-Time, and Memory-Augmented all reach $\Delta \approx -0.56$, compared with -0.221 for Standard. Additive and Token-Time are statistically equivalent on drift score (paired TOST, $\delta = 0.05$), and Memory-Augmented does not improve over Token-Time.

Implementation details. All models share the same Transformer encoder: $d = 64$, 4 attention heads, 2 layers, feed-forward width 128, dropout 0.1. The period encoding projects 32 sinusoidal features of t/T to d via a two-layer MLP. Training uses AdamW (lr $= 10^{-3}$, weight decay $= 10^{-2}$) with a cosine annealing schedule, 30 epochs, batch size 64. The corpus contains 5,000 SVO sentences; MLM training uses 3,159 sentences from periods $t0$–$t7$; 763 sentences from the same periods form the held-out evaluation set; 794 sentences from $t8$–$t9$ are reserved for the continuation diagnostic.

6.1 Temporal Neighborhood Movement

Figure 3 and Table 1 show the context drift score across periods. Standard is a sanity check: its movement ($\Delta = -0.221$, 95% CI $[-0.240, -0.203]$) is less than half the conditioned-model drift. All three conditioned models reach $\Delta \approx -0.56$, with overlapping confidence intervals; the gap between any conditioned model and Standard is the geometric evidence for the effect of temporal conditioning.

The class-specific curves serve as controls (Table 1). Stable subjects show smaller movement in absolute magnitude (Δ between -0.150 and -0.180 for conditioned models, versus $\Delta \approx -0.56$ for drifting subjects), but the direction is consistently negative. This reflects a global displacement of the embedding space induced by the period vector: even subjects with flat planted trajectories are pulled along the learned temporal direction, at lower magnitude. Bifurcating subjects show intermediate drift: conditioned models reach $t9$ values near 0.53, matching the planted endpoint $P(N_1 \mid s, t9) \approx 0.5$ for the mixed state. The geometric diagnostic distinguishes all three trajectory classes without relying on probe accuracy alone.

Paired equivalence testing (TOST, $\delta = 0.05$) formalizes this: diff $= -0.003$, 90% CI $[-0.026, +0.019]$, $p_{\mathrm{TOST}} < 0.001$; equivalence holds also at $\delta = 0.034$

Fig. 3. Context drift score across periods. All conditioned models move drifting subjects from N_1 toward N_2 more than Standard; Additive, Token-Time, and Memory-Augmented produce similar trajectories. Stable and bifurcating subjects provide class-specific controls.

($p_{\mathrm{TOST}} = 0.014$). D2 therefore functions as a sanity check that conditioning works, not as an arbiter between conditioning strategies.

6.2 Probe and Contrastive Diagnostics

The linear probe results reveal a distinction that drift score does not. On the standard (75%-fidelity) split, where local markers carry reliable context signal, all conditioned models outperform Standard (0.875 [$\pm$0.005]): Additive reaches 0.892 [$\pm$0.003], Token-Time 0.881 [$\pm$0.005], and Mem-Aug. 0.881 [$\pm$0.006]; on this split Additive is above Token-Time. The ordering reverses on the ambiguous split, where local verb and object markers are uninformative: Standard reaches 0.565 $[0.561, 0.568]$, Additive 0.579 $[0.575, 0.584]$, Token-Time 0.595 $[0.589, 0.601]$, and Memory-Augmented 0.598 $[0.590, 0.606]$. The confidence intervals for Additive and Token-Time do not overlap, showing a reproducible advantage for joint token-time conditioning when only subject identity and period remain as evidence. On the continuation split, Additive and Token-Time reach the same accuracy (0.763; Table 2), matching their overlap on drift score. Probe accuracy alone is insufficient to establish temporal traceability because it measures recoverability of the planted label, not neighborhood geometry. Even so, the split-dependent pattern suggests that joint conditioning encodes temporal information more precisely in the subject representation, with the advantage visible only when local markers are removed.

The contrastive sign-flip rates are similar across conditioned models: Standard 0.000, Additive 0.602 $[0.549, 0.655]$, Token-Time 0.624 $[0.576, 0.673]$, Mem-Aug. 0.587 $[0.536, 0.638]$. Confidence intervals overlap, so the contrastive diagnostic does not separate the conditioned models.

Table 1. Context drift score at the first and last periods. Values are means over 31 seeds; $\Delta = t9 - t0$; 95% CI half-widths in brackets.

Class	**Model**	**t0**	**t9**	$\boldsymbol{\Delta}$ [CI$_\pm$]
Stable	Standard	0.741	0.781	+0.040 [±0.017]
	Additive	0.827	0.678	−0.150 [±0.027]
	Token-Time	0.835	0.655	−0.180 [±0.024]
	Mem-Aug.	0.830	0.672	−0.158 [±0.021]
Drift	Standard	0.632	0.410	−0.221 [±0.018]
	Additive	0.839	0.272	−0.567 [±0.019]
	Token-Time	0.833	0.269	−0.563 [±0.020]
	Mem-Aug.	0.833	0.274	−0.559 [±0.022]
Bifurc.	Standard	0.793	0.660	−0.133 [±0.018]
	Additive	0.895	0.530	−0.365 [±0.017]
	Token-Time	0.891	0.525	−0.366 [±0.016]
	Mem-Aug.	0.878	0.530	−0.348 [±0.015]

Table 2. Main traceability diagnostics (means, 31 seeds). Drift: context drift score Δ for drifting subjects; Flip: contrastive sign-flip rate (D3); Ambig./Cont.: linear-probe accuracy on h_s. 95% CI in brackets.

Model	**Drift**	**Flip**	**Ambig.**	**Cont.**
Standard	−0.221 [±0.018]	0.000	0.565 [±0.004]	0.730 [±0.004]
Additive	−0.567 [±0.019]	0.602 [±0.053]	0.579 [±0.005]	0.763 [±0.003]
Token-Time	−0.563 [±0.020]	0.624 [±0.048]	0.595 [±0.006]	0.763 [±0.004]
Mem-Aug.	−0.559 [±0.022]	0.587 [±0.051]	0.598 [±0.008]	0.755 [±0.005]

6.3 Memory Diagnostics

The Memory-Augmented extension does not improve over Token-Time. Across 31 seeds, learned prototypes reach a continuation accuracy of 0.755 $[0.750, 0.760]$, marginally below Token-Time at 0.763 $[0.759, 0.766]$ ($\Delta = -0.008$). The oracle diagnostic shows the limits of this architecture: replacing learned prototypes with fixed vectors encoding the true $P(N_1 \mid s, t)$ trajectory yields a continuation of 0.769 $[0.761, 0.777]$ and a sign-flip rate of 0.694 $[0.576, 0.812]$, versus 0.624 for Token-Time. The sign-flip difference ($\Delta = +0.070$) exceeds the continuation improvement ($\Delta = +0.007$), but both CIs are wide. The MLM objective optimizes token prediction, not trajectory geometry. Perfect historical prototypes can improve contrastive behavior, which depends on local period sensitivity, but they do not replace a training signal for geometric continuity across unseen periods.

7 Discussion and Conclusion

No architecture matches the planted trajectories across all diagnostics. Conditioned models improve neighborhood drift, with $\Delta \approx -0.563$ against -0.221 for Standard, but even oracle memory raises continuation accuracy only to 0.769.

Table 3. Memory diagnostics (means, 31 seeds). Continuation (D4): linear-probe accuracy on held-out periods $t8$–$t9$; Flip: contrastive sign-flip rate (D3); 95% CI half-widths in brackets.

Model	Cont.	Flip	Δ cont. vs. Token-Time
Token-Time (ref.)	0.763 [±0.004]	0.624 [±0.048]	–
Mem-Aug. learned	0.755 [±0.005]	0.587 [±0.051]	−0.008 [±0.006]
Mem-Aug. oracle	0.769 [±0.004]	0.694 [±0.059]	+0.007 [±0.007]

This pattern points to the MLM objective, which rewards token prediction rather than geometric consistency across periods. Traceability appears only as a by-product of that objective.

The form of conditioning matters less than its presence. Paired equivalence testing (TOST, $\delta = 0.05$) confirms that Additive and Token-Time are equivalent on drift score; Token-Time shows a reproducible advantage only on the ambiguous probe (0.595 vs. 0.579, non-overlapping CIs), where local markers are uninformative. Layer-wise CKA provides a representational account: Additive vs. Token-Time similarity increases from 0.726 at the embedding layer to 0.887 (95% CI [0.856, 0.919]) at the final encoder layer, while Standard stays near 0.22, suggesting self-attention partially absorbs the input-level difference before it reaches the subject's final hidden state. The convergence of two architecturally distinct designs toward the same drift performance points to the training objective as the limiting factor.

The Memory-Augmented extension points to a second failure mode. For bifurcating subjects, which develop two coexisting senses in later periods, a mean prototype lies between clusters and represents neither well. The oracle diagnostic shows that the memory pathway can use better historical signals: fixed vectors encoding the true $P(N_1 \mid s, t)$ trajectory increase the sign-flip rate by +0.070 over Token-Time (0.694 vs. 0.624). The learned mean prototypes, however, cannot preserve the internal structure of a word's sense distribution. Geometry-based mode-separation metrics (silhouette, centroid distance, AUROC) confirm this: no architecture separates the two sense clusters better than Additive for bifurcating subjects, and Memory-Augmented does not improve over Token-Time.

These conclusions come from controlled diagnostics, not open-domain diachronic language. The probe-level advantage of Token-Time over Additive is statistically stable but modest in magnitude (1.6 percentage points) and does not appear on the drift score; it may not generalize beyond the controlled setting of fixed subject positions and small vocabulary. The diagnostic suite was co-designed with the architecture family, limiting its role as an independent evaluation for models outside this design space.

Two directions remain open. First, attention-level analysis is consistent with the layer-wise representational convergence noted above: Token-Time layer-0 heads show greater temporal variance in subject-directed attention mass (L0H3: 0.0075 vs. Additive 0.0033), yet ablating individual heads only partially reduces representational alignment (L0H0: $\Delta = -0.026$; L0H1: $\Delta = -0.021$), suggesting the effect is distributed rather than localized. A pilot noise sweep (3 seeds per fidelity level) is consistent with Token-Time retaining a drift advantage under degraded markers (slope= $+1.07$, $p = 0.002$), but replication with more seeds is needed before drawing conclusions. Second, memory mechanisms need to preserve the internal structure of historical representations rather than collapsing each period to one prototype, which remains the primary open problem for bifurcating subjects.

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